



SUBJECT : PROGRAMMING IN JAVA

SUBJECT CODE : CSI26W1

LEARNING UNIT: 7 (JAVA COLLECTION FRAMEWORK)

PROGRAMMING EXERCICES

Task 1:

Write a Java program that creates an `Integer ArrayList` and store at least five integers in it. Use an iterator to display all the numbers. Then prompt the user for a value to remove from the list. If the number appears in the list, remove it. Display the list again.

Save the file as `RemoveNumber.java`.

Task 2:

Write a Java program that creates a `String ArrayList`. Continually prompt the user for Strings to add to the list until the user enters a sentinel value. Display the Strings in a single line separated by commas.

Save the file as `StringArrayList.java`.

Task 3:

- a) John Doe has purchased buildings that contain apartments to rent. Create an `Apartment` class that contains fields for the street address, apartment number, monthly rent amount, and number of bedrooms for each `Apartment` that he owns. Include a constructor that assigns field values to the class, a `toString()` method that displays the `Apartment` values, and a `compareTo()` method that compares `Apartment` objects based on the rent value. Include any other methods that you think you might need.

Save the file as `Apartment.java`.

- b) Write an application that uses an array to store `Apartment` objects. Declare an array of 10 `Apartment` objects and prompt the user for their values. Then display the details of all apartments.

Save the file as ApartmentsArray.java

- c) Write an application that contains a `ArrayList` that stores `Apartment` objects. Prompt the user for values for `Apartments` and add them to the `LinkedList` until a sentinel value is entered. Sort the `Apartments` by rent value, and then display the sorted list.

Save the file as `ApartmentsArrayList.java`.

- d) Alter the application you wrote in c) above to use `LinkedList`. Sort the `Apartments` by number of rooms and then display the sorted list.

Save the file as `ApartmentsLinkedList.java`.

- e) John Doe sometimes sells one of his buildings. Modify the `ApartmentsLinkedList` application so that after data entry is complete and the list of `Apartment` objects is displayed, the application prompts the user for a street address to eliminate. Remove all the `Apartment` objects at the specified street address, and display the list of `Apartments` again.

Save the file as `ApartmentsLinkedListMod.java`.

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